

# How to Update RIOT Software on a Slocum Glider

Last Updated: June 23, 2026

## Instructions:

- 1) Download a pre-compiled MCUMGR from [GitHub](#).
- 2) Open a terminal window and cd to your downloaded MCUMGR files.

*Ex: C:\Program Files (x86)\mcumgr-client-windows-x86*

- 3) Download **zephyr-slocum.signed.bin.beta** from this [repo](#). Copy it to the same local directory as mcumgr-client and rename it to **zephyr-slocum.signed.bin**.
- 4) Connect the MCIL6F RIOT debug cable to the DEBUG bulkhead connector on the science bay stiffener ring and connect the USB-serial connector to your computer.
- 5) Open Tera Term (or equivalent terminal emulator) to baud rate 115200, 8-N-1. Hit <CR> and you should see the following prompt:

```
zaxy:~>
```

- 6) Power off the RIOT subsystem and issue the reboot command:

```
zaxy:~> riot schedule end
```

```
zaxy:~> kernel reboot cold
```

- 7) QUICKLY close Tera Term and navigate to the opened terminal window.
- 8) Probe the board. This will halt the boot process unless a mcumgr reboot is issued or the system is power cycled. **You have 2.5 seconds to issue the following command.** If the uController does not receive comms during a 2.5s window, boot will continue to the application, and you'll need to start again.

```
./mcumgr-client -d <serial-port> list
```

- 9) With the boot interrupted, upload the software to the board. You should see a progress bar if the upload is working properly.

```
./mcumgr-client -d <serial-port> upload zephyr-slocum.signed.bin
```

10) Reboot the board. This reboot will pass through the bootloader.

```
./mcumgr-client -d <serial-port> reset
```

11) After issuing this command, you can open Tera Term to **<serial-port>** and should see zaxy boot as normal after 2.5 seconds. A hard power cycle could also be used here by turning the glider off, removing the green plug, and reconnecting the green plug again to reboot the glider.